CSI 2101: Discrete Structures

2. PROOFS

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reasoning about quantifiers

(1) Every even number is the sum of two odd numbers.

$2 = 1 + 1$

$4 = 3 + 1$

$0 = (-1) + 1$

TRUE BECAUSE

$$\underset{\text{EVEN}}{n} = \underset{\text{ODD}}{(n-1)} + \underset{\text{ODD}}{1}$$

(2) For every $n \geq 0$, $n^2 + n + 41$ is a prime.

n	n^2+n+41	
0	41	✓
1	43	✓
2	47	✓
5	71	✓
10	151	✓

FALSE!

$n = 41$

$41^2 + 41 + 41 = 41(41+1+1)$

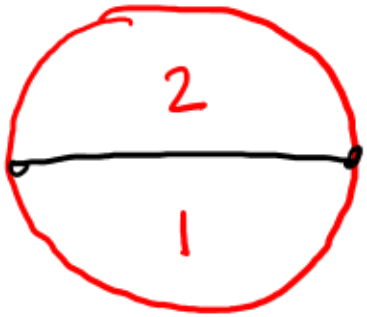
$= 41 \cdot 43$

IS A COMPOSITE.

primes

2	3	5	7	11	13	17	19	23	29
31	37	41	43	47	53	59	61	67	71
73	79	83	89	97	101	103	107	109	113
127	131	137	139	149	151	157	163	167	173

(3) Inscribe a regular n-gon on a circle and draw all diagonals. How many regions are there?



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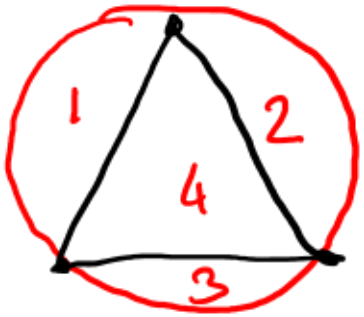

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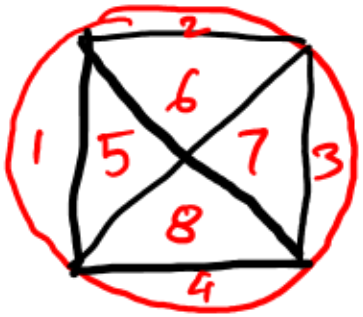
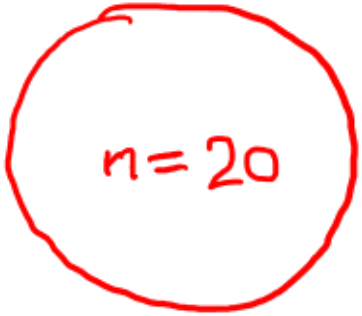


8

16

30

≤ 31 EVEN IF IRREGULAR



$2^{19} = 524288$

ACTUAL $= 5036$

(4a) 29 is the sum of three integer cubes.

$$29 = 1^3 + 1^3 + 3^3$$

(4b) 33 is the sum of three integer cubes.

$$\begin{aligned} 33 = {} & 8866128975287528^3 \\ & + (-8778405442862239)^3 \\ & + (-2736111468807040)^3 \end{aligned}$$

A. Booker, *Cracking the problem with 33*, 2022

(5) Every even number greater than 2 is the sum of two primes.

$4 = 2 + 2$

$6 = 3 + 3$

$8 = 3 + 5$

$10 = 3 + 7$ OR $5 + 5$

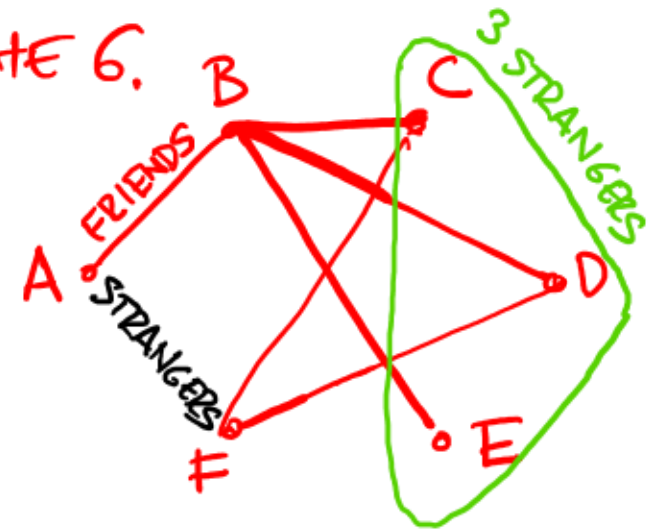
$100 = 47 + 53$ OR $97 + 3$ OR $89 + 11$ OR $83 + 17$...

GOLDBACH'S CONJECTURE (1742)

UNSOLVED

Theorem: Among any 6 people there is a group of 3 friends or a group of 3 strangers.

Proof: LET Alice BE ONE OF THE 6.
THERE ARE 5 OTHERS, SO
Alice IS FRIENDS WITH ≥3 <u>OR</u>
Alice IS A STRANGER TO ≥3



<u>CASE 1.</u> Alice HAS ≥3 FRIENDS.
CALL THEM F.

<u>SUBCASE 1.</u> NO FRIENDS WITHIN F.
THEN F IS A GROUP OF 3 STRANGERS.

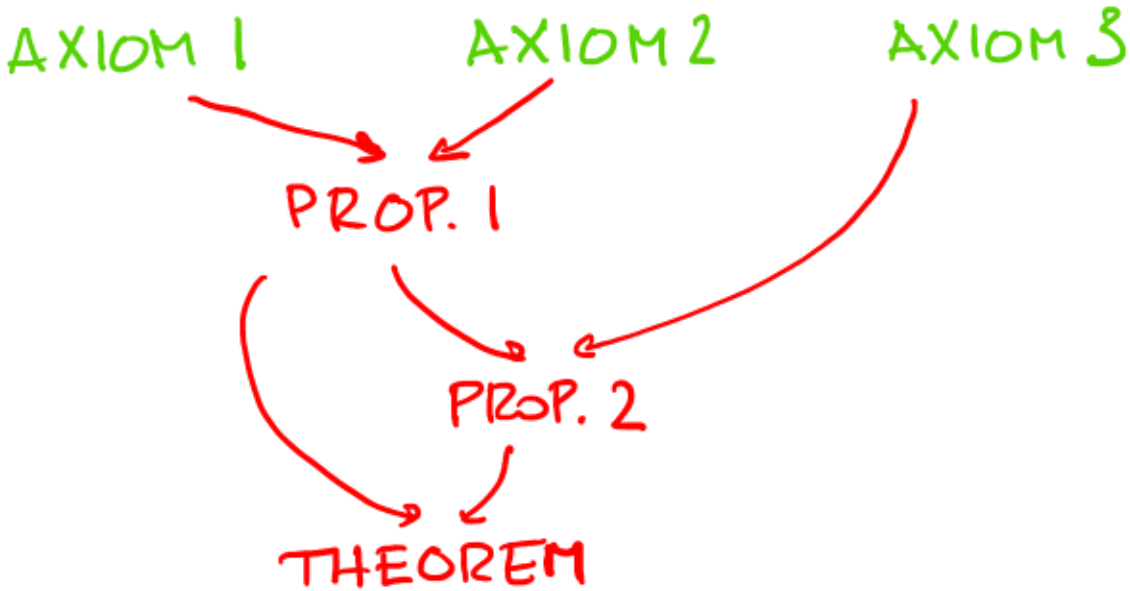
<u>SUBCASE 2.</u> THERE ARE FRIENDS WITHIN F.
CALL THEM x AND y.
THEN {Alice, x, y} IS A GROUP OF 3 FRIENDS.

<u>CASE 2</u>. Alice IS STRANGER TO ≥ 3 OTHERS. CALL THEM S.

<u>SUBCASE 1</u>. EVERY PAIR WITHIN S ARE FRIENDS. THEN S IS A GROUP OF 3 FRIENDS.

<u>SUBCASE 2</u>. SOME PAIR x, y WITHIN S ARE STRANGERS. THEN {Alice, x, y} IS A GROUP OF 3 STRANGERS. $\square$

A proof is a sequence of deductions from axioms and already proved theorems that concludes with the proposition we want.



A theorem is a proposition that we (or someone else) already proved.

axioms

Axiom 1: If Alice is friends with Bob then Bob is friends with Alice

Axiom 2: If Alice is in a group of 6 people then there are 5 others

proof by cases

THE CASES MUST COVER ALL POSSIBILITIES

$$\frac{C_1 \text{ OR } C_2 \qquad C_1 \to P \qquad C_2 \to P}{P}$$

C_1	C_2	P	$\overbrace{C_1 \text{ OR } C_2}^{A}$	$\overbrace{C_1 \to P}^{B}$	$\overbrace{C_2 \to P}^{C}$	$\overbrace{A \text{ AND } B \text{ AND } C}^{D}$	$D \to P$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
				⋮			⋮
F	F	F	F	T	T	F	T

ALWAYS T
PROOF METHOD
IS <u>SOUND</u>

Proofs are easy to verify* but can be difficult to create.

There are famous propositions waiting for proofs even after 100+ years.
They are called conjectures.

proving an implication

Theorem: The sum of even integers is even.

x EVEN, y EVEN $\longrightarrow$ $x+y$ EVEN

Proof: ASSUME x, y ARE EVEN.

$x = 2 \cdot m$ FOR SOME m

$y = 2 \cdot n$ FOR SOME n

$x+y = 2m + 2n = 2(m+n)$ IS TWICE AN INTEGER SO IT IS EVEN. $\square$

Theorem: The product of odd integers is odd.

Proof: ASSUME x, y ARE ODD.

$x = 2 \cdot m + 1$ FOR SOME m

$y = 2 \cdot n + 1$ FOR SOME n

$$x \cdot y = (2m+1)(2n+1)$$
$$= 4mn + 2m + 2n + 1$$
$$= 2(2mn + m + n) + 1$$

THIS IS OF THE FORM $2k+1$ WHICH IS ODD. $\square$

Theorem: The square of an odd number is of the form $8k + 1$ for some integer k.

Proof: ASSUME $n = 2p+1$.

$$n^2 = (2p+1)^2$$
$$= 4p^2 + 4p + 1$$
$$= 4(p^2+p)+1$$

n	n^2
1	$1 = 8\cdot 0 + 1$
3	$9 = 8\cdot 1 + 1$
5	$25 = 8\cdot 3 + 1$

NICE IF IT WAS EVEN..

<u>Conjecture</u>: p^2+p IS ALWAYS EVEN.

p	p^2+p
0	0 ✓
1	2 ✓
2	6 ✓
3	12 ✓

Proof?
- p EVEN $\rightarrow p^2+p =$ EVEN + EVEN
- p ODD $\rightarrow p^2+p =$ ODD + ODD

<u>Proof.</u> ASSUME n IS ODD SO $n = 2p+1$ FOR SOME INTEGER p. THEN

$$n^2 = (2p+1)^2 = 4p^2 + 4p + 1 = 4p(p+1) + 1.$$

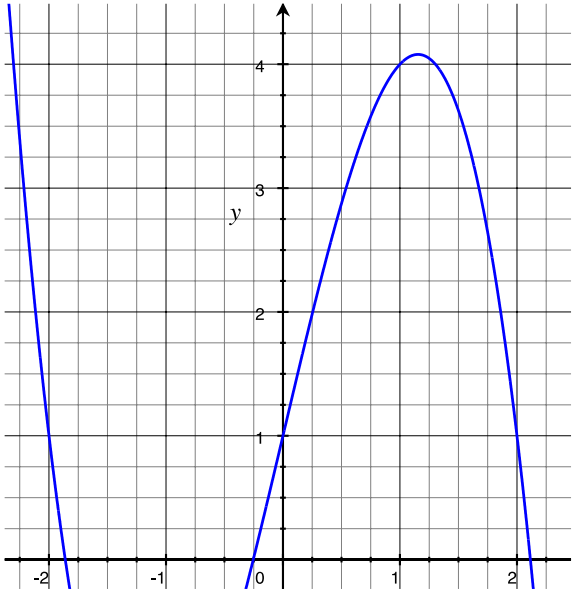
CASE 1: p IS EVEN. $p = 2t$ FOR SOME t.

THEN $n^2 = 4 \cdot 2t(p+1) + 1 = 8t(p+1) + 1$ IS OF THE FORM $8k+1$.

CASE 2: p IS ODD. $p = 2t+1$ FOR SOME t.

THEN $n^2 = 4p(2t+2) + 1 = 8p(t+1) + 1$ IS ALSO OF THE FORM $8 \cdot k + 1$. $\square$

Theorem: If $0 \leq x \leq 2$, then $-x^3 + 4x + 1 > 0$.



proof by contrapositive

Theorem: If $r \geq 0$ is irrational then $\sqrt{r}$ is irrational.

Lemma: In every group of six people including Alice, she has 3 friends or is stranger to some 3.

proving equivalences

Theorem: n^2 is even if and only if n is even.

proof by contradiction

Theorem: $\sqrt{2}$ is irrational.

A table is filled with distinct numbers. A saddle is a number largest in its column and smallest in its row.

Theorem: A table can have at most one saddle.

A table is filled with distinct numbers. Sort the rows. Then sort the columns. Are the rows still sorted?